Natural Language Processing Course Project

Phase 1 Project Proposal Track C: Research Team 12

1. Project Title

Evidence Deduplication and Selective Answering in Retrieval-Augmented QA under Incomplete Source-Dependency Information

2. Team Members & Responsibilities

Zhou Yu: Literature review; research-question refinement; controlled dataset and source-lineage labels; annotation guidance and related work. Dataset design will be cross-reviewed by Wang Ce.

Wang Ce: Experimental environment; baseline and proposed-method implementation; model/version and cost tracking; code and run documentation. The method will be reviewed by Xue Jian.

Cheng Zihan: Evaluation metrics; experiment and ablation scheduling; paired bootstrap and risk-coverage analysis; results visualization. Metric definitions will be reviewed by Zhou Yu.

Yang Haoran: Experiment execution and raw-output logs; QACC/GroupQA external checks where feasible; error analysis and example audit. Case interpretations will be reviewed by Wang Ce.

Xue Jian: Proposal and report coordination; ACL-style final report and slides; timeline and individual-contribution records. Zhou Yu will review the integrated report for consistency.

All five members will discuss the hypotheses, review key results, and participate in the final presentation. Actual contributions will be recorded as the project proceeds.

3. Background & Motivation

Retrieval-augmented question answering (RAG) gives a language model documents to support its answer. Several retrieved pages may repeat or paraphrase one original source, so repeated claims do not necessarily represent independent confirmation. At the same time, independent sources can provide valuable support even when their wording is similar. Text similarity alone therefore cannot reliably identify source dependence.

Prior work studies conflicting contexts (QACC), evidence sufficiency and abstention (Sufficient Context), repetition effects (GroupQA), and evidence arbitration (ArbGraph) [1–4]. This project examines the reliability of these strategies when source relationships are missing or incorrect. We will separate source lineage from semantic similarity and test both benefits and failure cases, without assuming that deduplication improves accuracy.

4. Problem Definition / Research Questions

Given a question, a fixed set of retrieved passages, and possibly incomplete or noisy source metadata, the system must provide a short answer with supporting evidence or abstain when the evidence is insufficient or cannot support a reliable answer. We will study the generation stage with retrieval held fixed. Source-lineage ground truth will be used for evaluation and a clearly separated oracle comparison, not as an input to the practical method.

RQ1: Does repeating evidence from one source change answer accuracy or the decision to answer? RQ2: How do missing or incorrect source labels affect exact/semantic deduplication and selective answering? RQ3: Can a lightweight method account for source-dependency uncertainty while retaining useful evidence from independent sources?

We hypothesize that repetition affects some model decisions; that source-aware aggregation with an explicit unknown state may improve risk at matched answer coverage over semantic deduplication alone; and that aggressive deduplication may remove useful independent evidence. These hypotheses may be rejected by the results.

We will build paired controlled questions with same-source copies or paraphrases and similar claims from independent sources, under complete, missing, and deliberately corrupted metadata. A small pilot will check construction and cost; if feasible, we will evaluate 60 base questions (20 development, 40 test), keeping all variants of each question in one split. Cases will cover sufficient consistent evidence, identifiable errors, unresolved conflict, and missing key evidence. We will compare direct answering, an abstention prompt, exact and semantic deduplication, sufficiency/confidence gating, a combined baseline, and our lightweight

source-aware aggregation and gating method. One frozen model is the minimum; a second model and QACC/GroupQA external checks are optional subject to resources.

We will report exact match/F1, coverage, selective risk at matched coverage, abstention errors, independent-support retention, citation support, and call cost. Thresholds will be set on development data. Paired bootstrap confidence intervals will resample base questions, not their variants. We will also test whether results change under natural versus length-controlled contexts.

5. Expected Outcomes

Expected deliverables are a documented, reproducible controlled evaluation protocol; implementations of the selected baselines and lightweight proposed method; experimental results with error and ablation analyses; and run instructions for the course project. The study should clarify when source metadata helps, when it misleads, and when deduplication removes useful evidence. Performance gains are an empirical question, not a promised outcome. The pilot and fixed-model design keep the work feasible without training a large model.

6. Project Timeline

Period	**Planned work**
Oct 10–11	A: literature and dataset draft; B: environment setup; C: metric outline; D: sample feasibility check; E: proposal draft and coordination.
By Oct 12	E integrates the 1–2 page proposal; all members review and confirm responsibilities and study design.
Week 1 after proposal	A: MVP data and labels; B: environment and baselines B0–B5; C: core metric scripts; D: pilot; E: midterm outline.
Week 2 / midterm milestone (date TBD)	B: implement method B7; C: schedule main evaluation; D: collect preliminary outputs; A: refine labels; E: prepare milestone report and slides.
Weeks 3–4	Run main and ablation experiments plus feasible QACC/GroupQA checks; complete paired analysis, visualizations, error cases, and related work.
Weeks 5–6 / final deadline (TBD)	Integrate the ACL-style report, code and run guide, contribution statement, and final slides; all members review the submitted materials.

Note: the instructor has not announced the midterm or final submission dates. These later tasks are scheduled relative to the proposal date and will be adjusted when the official dates are released.

References

[1] Liu, S. et al. 2025. Open Domain Question Answering with Conflicting Contexts. Findings of NAACL 2025. https://aclanthology.org/2025.findings-naacl.99/

[2] Joren, H. et al. 2025. Sufficient Context: A New Lens on Retrieval Augmented Generation Systems. ICLR 2025. https://arxiv.org/abs/2411.06037

[3] Naphade, A. 2026. Rational Synthesizers or Heuristic Followers? Analyzing LLMs in RAG-based Question-Answering. Findings of ACL 2026. https://aclanthology.org/2026.findings-acl.2003/

[4] Niu, Q. et al. 2026. ArbGraph: Conflict-Aware Evidence Arbitration for Reliable Long-Form Retrieval-Augmented Generation. arXiv:2604.18362. https://arxiv.org/abs/2604.18362